



Properties of Definite Integrals

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Apply the constant multiple and sum/difference properties of definite integrals
- Use the property of reversing limits of integration with a sign change
- Combine adjacent integration intervals using the additivity property

Use the given integral values and the properties of definite integrals to evaluate each expression below.

1. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_0^a [2f(x) + 3g(x)] dx$$

Answer: _____

2. Given that the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_4^1 f(x) dx$$

Answer: _____

3. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_{-1}^4 f(x) dx$$

Answer: _____

4. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_0^a [5f(x) - 2g(x)] dx$$

Answer: _____

5. Given that the integral of $h(x)$ from -1 to 2 equals 7, evaluate the following.

$$\int_2^{-1} 4h(x) dx$$

Answer: _____



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6. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_4^{-1} f(x) dx$$

Answer: _____

7. Given that the integral of $f(x)$ from 0 to a equals 37, evaluate the following.

$$\int_0^a \frac{1}{2} f(x) dx$$

Answer: _____

8. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $h(x)$ from -1 to 2 equals 7, evaluate the following.

$$\int_{-1}^1 f(x) dx + \int_{-1}^2 2h(x) dx$$

Answer: _____

9. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_{-1}^4 3f(x) dx$$

Answer: _____

10. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_a^0 [f(x) + g(x)] dx$$

Answer: _____





Remind students that constants can be factored out of integrals, swapping limits introduces a negative sign, and adjacent intervals can be combined via additivity.

Solutions

1. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_0^a [2f(x) + 3g(x)] dx$$

- Split the integral into two separate integrals using the sum property.
- Factor the constants 2 and 3 outside their respective integrals.
- Substitute 37 for the integral of $f(x)$ and 16 for the integral of $g(x)$.
- Compute 2 times 37 plus 3 times 16 to get 122.

Answer: 122

2. Given that the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_4^1 f(x) dx$$

- Notice the lower limit is greater than the upper limit.
- Apply the reversal property by swapping the limits and introducing a negative sign.
- Replace the integral from 1 to 4 with -2.
- Multiply negative one by negative two to obtain positive two.

Answer: 2

3. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_{-1}^4 f(x) dx$$

- Use the additivity property to split the interval at x equals 1.
- Write the integral as the integral from -1 to 1 plus the integral from 1 to 4.
- Substitute 5 and -2 for the two integrals.
- Add 5 and -2 to get 3.

Answer: 3

4. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_0^a [5f(x) - 2g(x)] dx$$

- Apply the difference property to split the integral into two pieces.
- Pull the constants 5 and -2 outside the integrals.
- Substitute 37 for the integral of $f(x)$ and 16 for the integral of $g(x)$.
- Compute 5 times 37 minus 2 times 16 to get 185 minus 32 equals 153.

Answer: 153



5. Given that the integral of $h(x)$ from -1 to 2 equals 7, evaluate the following.

$$\int_2^{-1} 4h(x) dx$$

- Factor the constant 4 outside the integral.
- Swap the limits of integration and introduce a negative sign.
- Replace the integral from -1 to 2 with 7.
- Compute -4 times 7 to get -28.

Answer: -28

6. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_4^{-1} f(x) dx$$

- Reverse the limits of integration to write the integral from -1 to 4 with a negative sign.
- Split the integral from -1 to 4 at x equals 1 using additivity.
- Substitute 5 and -2 for the two component integrals.
- The sum is 3, and applying the negative sign gives -3.

Answer: -3

7. Given that the integral of $f(x)$ from 0 to a equals 37, evaluate the following.

$$\int_0^a \frac{1}{2} f(x) dx$$

- Factor the constant one-half outside the integral.
- Substitute 37 for the integral of $f(x)$ from 0 to a .
- Multiply one-half by 37 to obtain thirty-seven halves.

Answer: $\frac{37}{2}$

8. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $h(x)$ from -1 to 2 equals 7, evaluate the following.

$$\int_{-1}^1 f(x) dx + \int_{-1}^2 2h(x) dx$$

- Evaluate the first integral by substituting its given value of 5.
- Factor the constant 2 outside the second integral.
- Substitute 7 for the integral of $h(x)$ and compute 2 times 7 equals 14.
- Add 5 and 14 to get 19.

Answer: 19

9. Given that the integral of $f(x)$ from -1 to 1 equals 5 and the integral of $f(x)$ from 1 to 4 equals -2, evaluate the following.

$$\int_{-1}^4 3f(x) dx$$

- Pull the constant 3 outside the integral.
- Use additivity to combine the intervals from -1 to 1 and from 1 to 4.
- Substitute 5 and -2 to get a combined integral value of 3.
- Multiply 3 by 3 to obtain 9.

Answer: 9



10. Given that the integral of $f(x)$ from 0 to a equals 37 and the integral of $g(x)$ from 0 to a equals 16, evaluate the following.

$$\int_a^0 [f(x) + g(x)] dx$$

- Reverse the limits of integration and introduce a negative sign.
- Split the resulting integral into the sum of two integrals from 0 to a .
- Substitute 37 for the integral of $f(x)$ and 16 for the integral of $g(x)$.
- Add 37 and 16 to get 53, then apply the negative sign to obtain -53.

Answer: -53

